UC San Diego

Power System Resource Adequacy in the Low-Carbon Transition
低碳转型中的电力系统资源充足性

—— Northeast China Grid Winter Resource Adequacy（中国东北电网冬季资源充足性研究）

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Background

Resource Adequacy（资源充足性）

Resource Adequacy refers to the ability of a power system to consistently meet the electricity demands of consumers by having sufficient capacity.

Most North American systems use “one day in ten year” loss of load expectation principle as the reliability standard for resource adequacy planning.*

Other international standards: Expected Unserved Energy (EUE) by NERC, Loss of Load Probability (LOLP) in Scandinavia, and Unserved Energy (USE) in Australia’s National Energy Market (NEM).

A Chinese equivalent is provincial FYP energy development plans that balance power generation and consumption, which serve as the primary mechanism for resource adequacy planning.

资源充足性指电力系统具备足够能力，持续满足消费者的用电需求。

大多数北美的电力系统使用“十年的一天”负荷缺电预期作为资源充足性规划中的可靠性标准。

国际上其他的常用标准：NERC的电力不足期望，北欧的供能不足概率，澳大利亚国家电力市场的未供给能量。

中国的类似的标准只有各省在五年计划里提到的电力生产和消耗平衡，这是资源充足性规划的主要体现。

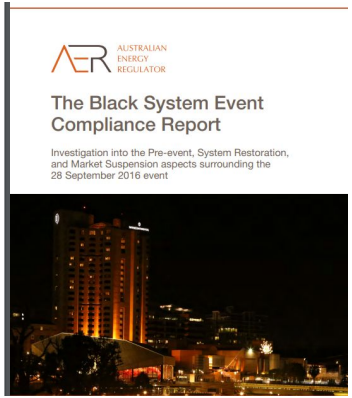
* The 1-in-10 Resource Adequacy Standard mean that planning reserve margins need to be high enough that involuntary load shedding due to inadequate supply would occur only once in ten years. See https://www.ferc.gov/sites/default/files/2020-05/02-07-14-consultant-report.pdf

Background

Routine Public Reports on Resource Adequacy（关于资源充足性的常规公开报告类型）

- Two types of routine public reports are important for supporting a reliable power system:
  - Post-crisis analysis reports (停电事故事后分析 报告)
    - China's late 2021 power crisis in the Northeast Grid (NEG)
    - 2021年底东北电网停电事故事后分析
  - Forward-looking seasonal resource adequacy outlook reports (季节性资源充足性展望 报告)
    - Resource adequacy outlook for Winter 2022/2023
    - 2022/2023年冬季资源充足性展望

- International Examples



Background

NEA (2023): Managing Power System Operation Risks
(关于完善电力系统运行方式分析制度 强化电力系统运行安全风险管控的通知)

- On Feb 2023, NEA issued a new policy document, “Notification on refining analysis on power system operation and enhancing management of power system operation risks”.
  - First policy document that regularized system-level forward-looking analysis to manage operation risks.
  - The policy document requires provincial grid companies to conduct the following analyses each year:
    - Post-crisis analysis
    - Annual forward-looking operation analysis
    - Summer & winter seasonal forward-looking analysis

- 能源局在2023年2月发布了《关于完善电力系统运行方式分析制度强化电力系统运行安全风险管控的通知》
  - 这是首份从系 统层面，规定进行常规化运行方式分析以 规避系统风险的文件。
  - 要求省级电网企业每年进行：
    - 事后电力安全事故情况分析
    - 系统年度运行分析
    - 迎峰度夏（冬）分析

Background

This Study (本研究特点)

- This research focused on the Northeast Grid, provided examples by conducting
  - Post-crisis analysis on the 2021 Northeast power crisis
  - 2022/2023 winter seasonal resource adequacy outlooks
- International collaborative team: UC San Diego, Regulatory Assistance Project, NCEPU
- Working paper version (contact for copy)
- All data and models will be available for use by the community once finalized

- 本研究以东北电网为主，进行以下两种分析以供参考：
  - “事后电力安全事故情况分析 ” <-> 2021年底东北电网停电事故事后分析
  - “迎峰度冬”分析 <-> 2022/2023年冬季资源充足性展望
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  - 华北电力大学：袁家海，张健，张凯
- 工作论文（请联系获取）
- 所有数据和模型将在完成后公开使用

Fall 2021 Post-crisis

Overview and Impact（概述与影响）

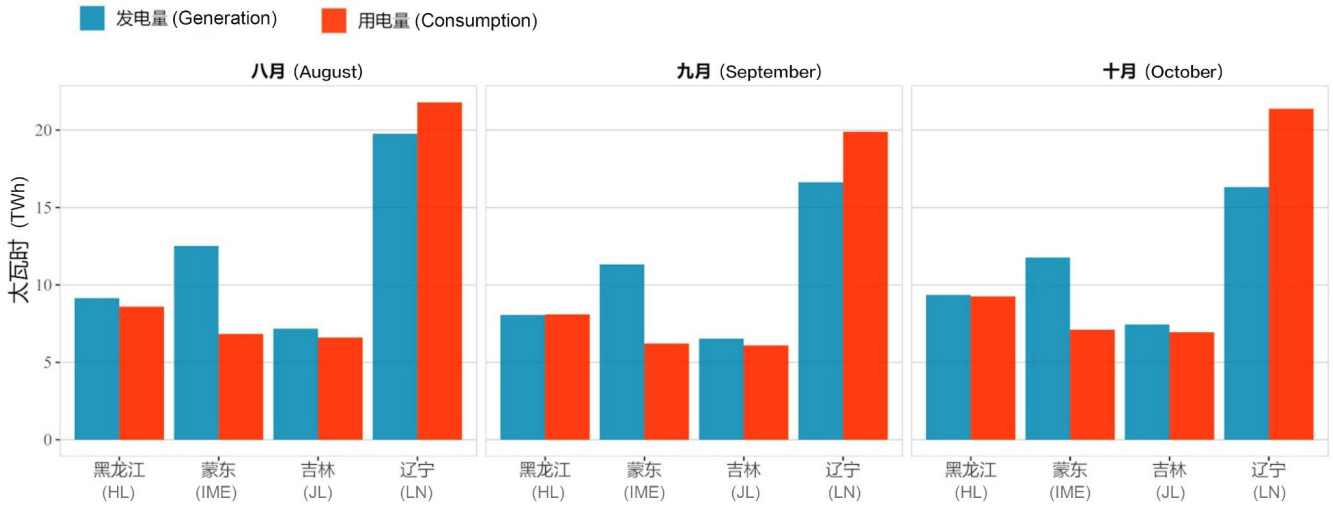
- In fall 2021, China experienced a widespread electricity crisis, with blackouts, electricity rationing, and orderly power consumption reductions implemented in around 20 provinces. Northeast China, including Heilongjiang, Jilin, Liaoning, Inner Mongolia East, was hit the hardest.
- Northeast China's power crisis led to unannounced power cuts, affecting transportation, water supply, and other basic social power consumption.
- Several potential factors have been evaluated:
  - Economic recovery and power demand surge
  - Rising coal prices and constrained supply
  - Contractual obligations for inter-regional exports
  - Effects of on-grid coal power price mechanism
  - Wind power output behavior

- 2021年秋季，中国经历了一场大范围的“电荒”，大约有20个省份出现了有序用电和拉闸限电等情况。东北地区包括黑龙江、吉林、辽宁和内蒙古东部地区，受到的影响最大。
- 东北“电荒”导致未经通知的停电，影响了交通、供水和其他基本社会用电。
- 多个潜在因素已经得到评估，包括：
  - 经济复苏和电力需求激增
  - 煤价上涨和供应受限
  - 跨区送电的合同义务
  - 上网煤电价格机制的影响
  - 风电的处理表现

Fall 2021 Post-crisis

Northeast Grid Supply and Demand, Fall 2021（2021年秋东北电网电力供应及需求）

- NEG region electricity generation and consumption during power shortage events of Fall 2021
  - Liaoning is the largest demand center and a net importer from the other provinces
  - Inner Mongolia East has a large generation surplus



- 辽宁是最大的电力消耗省和最大的净输入电力省。
- 蒙东电网的有最多的净发电盈余。

Fall 2021 Post-crisis

Key Narratives（主要因素）

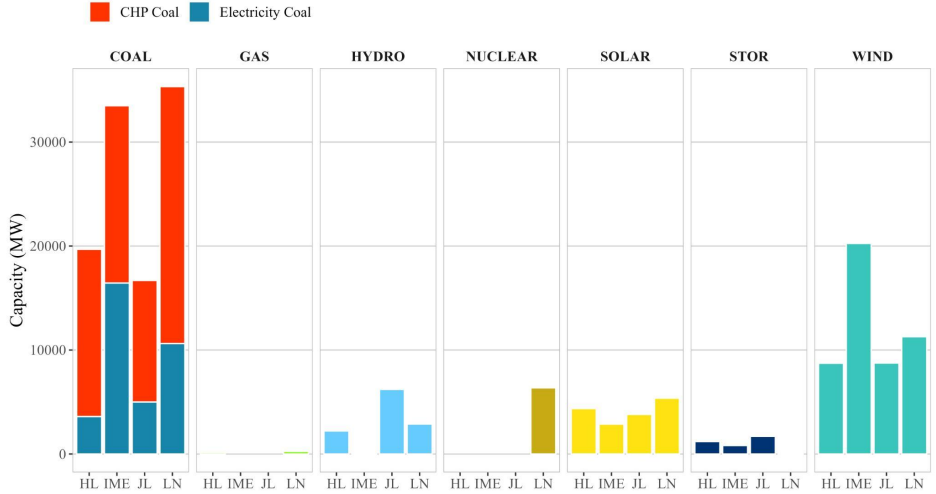
- The mounting electricity demand resulting from a strong economic recovery after COVID was a great contributor to the power crunch.
- Coal prices rose throughout the year, leading to an unexpectedly high thermal coal price and low coal inventories in many power plants throughout 2021. In the case of electricity price caps, power plants are unwilling to generate electricity at a loss and choose to shut down for maintenance.
- The northeast grid has long-term contractual obligations to export power to North China Grid, leading to inflexible scheduled flows over these lines.
- Wind power performance was not a main factor, because it has a small share in Liaoning, and the blackout happened before the worsened wind condition.

- 新冠疫情大流行过后强劲的经济复苏导致的电力需求激增。
- 煤价在全年持续上涨，导致2021年许多电厂的动力煤价格飙升，库存不足。在存在电价上限的情况下，电厂不愿亏本发电，选择停机检修，导致煤电供应不足。
- 东北电网长期负有向华北电网出口电力的合同义务，导致这些输电线路上僵化的持续外送电力，即使是在电荒发生时亦是如此。
- 风电表现不是这次危机的主要因素，因为它在辽宁占据较小的份额，并且此次电荒发生在风况恶化之前。

Model simulation for winter 2022/2023

Methodology(模型及方法论)

- **Objective:** Minimize production costs in NEG under various scenarios for December 2022 and February 2023.
- **Model:** Unit Commitment and Economic Dispatch (UCED)。
- **Model input:**
  - **Load:** Hourly provincial loads and NCG export profiles.
  - **Generation:** Installed capacity (below) from Global Energy Monitor and manual checking
  - **Transmission:** Interprovincial lines at 220-500kV with losses.



- **目标**:2022年12月和2023年2月，东北电网在不同情景设定下的最小发电成本。
- **模型**:机组组合与经济调度模型(UCED)。
- **模型输入**:
  - **负荷**:每小时的东北电网省级负荷，以及外送 华北的电量。
  - **发电**:装机容量(左 图)来自Global Energy Monitor以及手动筛查。
  - **输电**:考虑网损的220-500kV的跨省 输电线路。

Model simulation for winter 2022/2023

Scenarios（情景设置）

- **Baseline scenario** assumes normal operations (in generators and weather) and efficient spot market dispatch.
- Other scenarios explore the effects of existing policies and dispatch rules currently in use in NEG.
- All CHP units are guaranteed to stay committed in all scenarios.

Scenarios	Demand	Coal-Fired Capacity	Coal Prices	Renewable Profiles	NSE* Penalty in NCG
Extreme Weather 极端天气	+ 15%	-	-	Low availability	-
High Fuel Price and Derated Capacity 高燃料价格和缩减容量	-	Derated	High	-	-
Prioritized Export 优先外送	-	-	-	-	High
Combined 因素叠加	+15%	Derated	High	Low availability	High

- **基准情景**假设通常天气与发电厂运行下在高效开放市场下的运行情况。
- 其他情景（见左表）探讨了东北电网 各种现有政策和调度规则的影响。
- 在所有情景下，所有热电联产机组均保持投入。

Non-served energy (NSE) Penalty: *Produced when unfulfilled demand occurs during simulation at a considerably higher cost.*

Model simulation for winter 2022/2023

Results（仿真结果）

- The system costs for December 2022 and February 2023 under each scenario are shown in table below.
- The **Baseline scenario** has the lowest variable costs and the highest start costs, indicating efficient coal-fired generation with more frequent unit start-ups and shut-downs.
- In the **Extreme Weather scenario**, increased demand and limited availability of renewable energy result in higher coal-fired generation and reduced unit start-ups and shut-downs.
- In the **Combined scenario**, resource inadequacy results in minimized start-ups and shut-downs and high power generation from coal-fired plants, reflected in the highest variable costs and lowest start costs in both months.

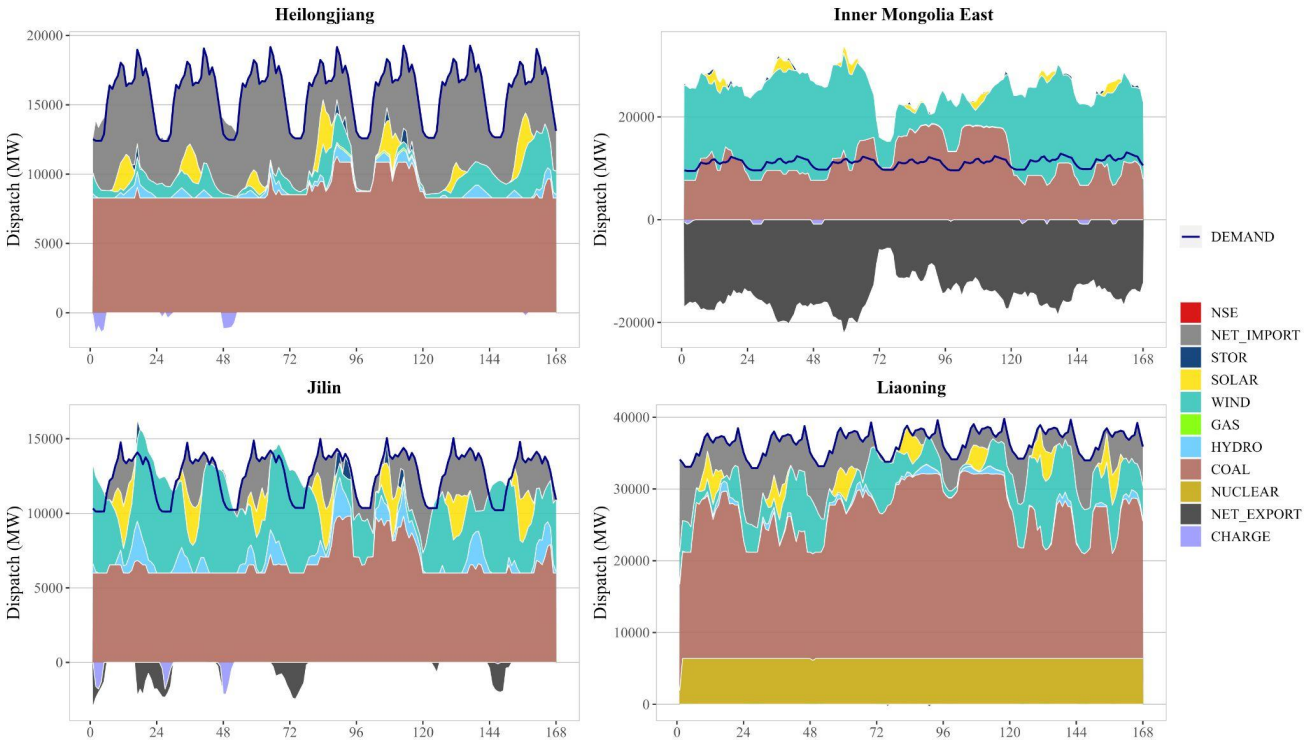
Unit: US$M **Exchange Rate:** 1 USD= 7.00 RMB	**Baseline**		**Extreme Weather**		**Combined**	
	December 2022	**February 2023**	**December 2022**	**February 2023**	**December 2022**	**February 2023**
Variable Cost	1,588	1,596	2,179	2,210	2,707	3,021
Start Cost	2.9	10.6	1.7	8.9	0	3.6

- **基准情景**有着最少的可变成本以及启动成本，意味着煤电机组经济地分配他们的发电资源。
- 在**极端天气情景**下，电力需求及可再生能源负荷系数降低，导致火电机组启停减少。
- **因素叠加情景**中，资源不充足导致最多的火力发电以及最少的机组启停。

Model simulation for winter 2022/2023

Results（仿真结果-2022年12月）

- The **Baseline scenario** maintains resource adequacy and fulfills export obligations to NCG.



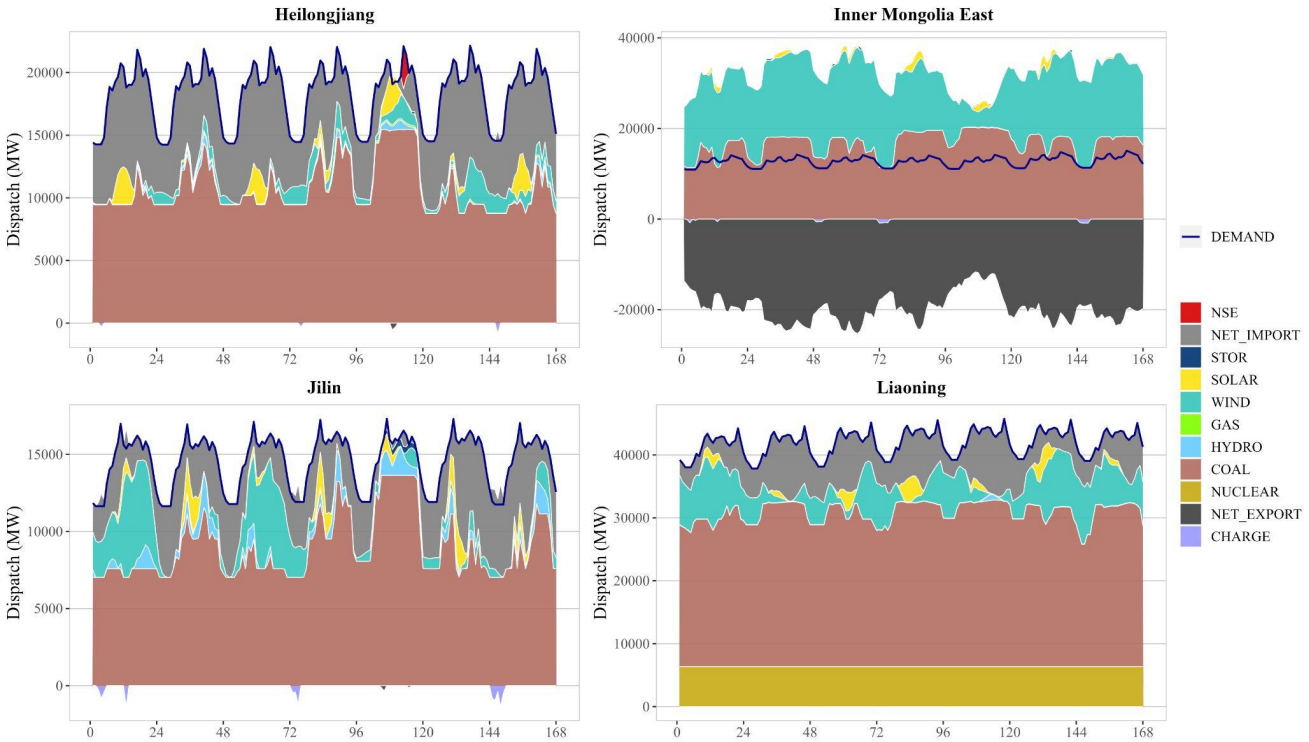
- Energy storage is charged at night when wind is abundant and discharged during peak hours to relieve stress on the system.

- 东北电网在**基准情景**下始终保持资源充足性，同时可以满足外输华北电网的合同义务。
- 储能设备在夜间风力资源充足时充电，高峰时段放电，以缓解系统压力

Model simulation for winter 2022/2023

Results（仿真结果-2022年12月）

- The **Extreme Weather scenario** maintain resource adequacy most of the time, with minor periods of NSE during peak hours.



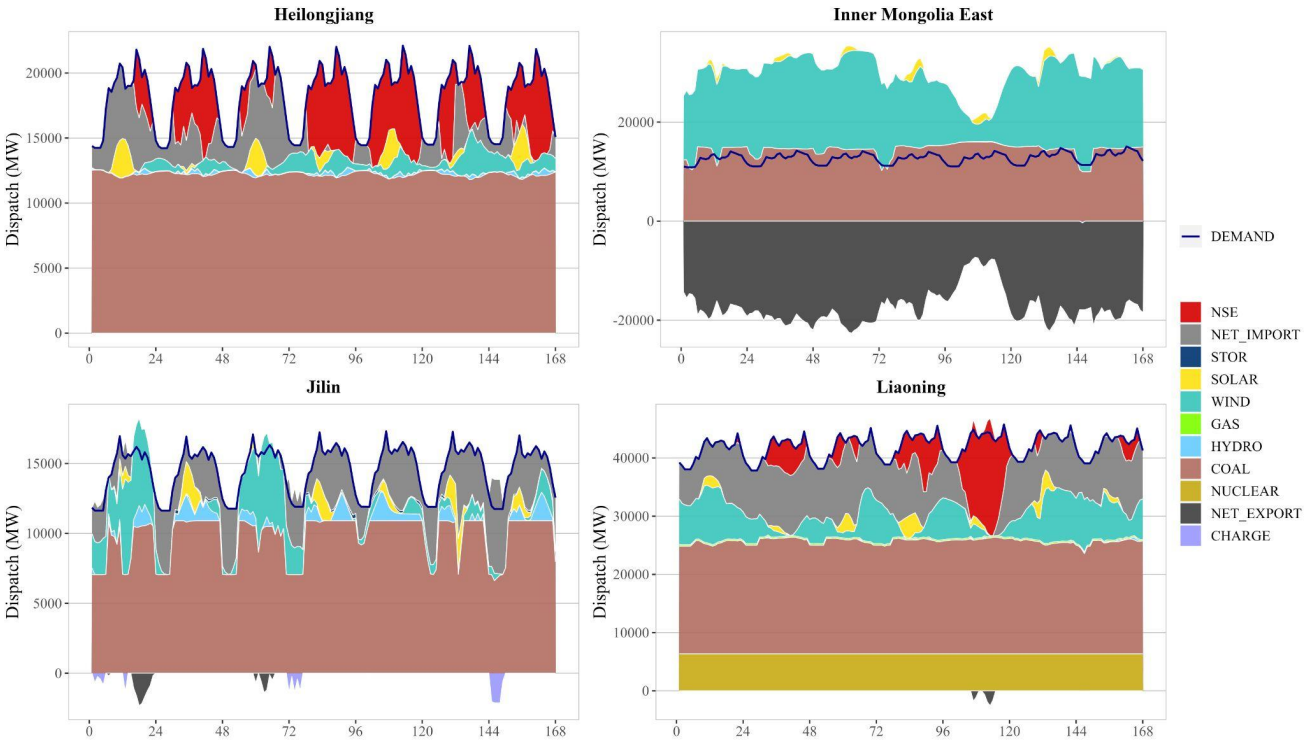
- Decrease in wind and increase in demand result in reduced amplitude and frequency of storage unit charging and discharging.

- 东北电网在**极端天气情景**下的绝大部分时间保持资源充足，在高峰时段有少量 NSE。
- 风电资源减少与电力需求增加导致储能设备充放电频率降低。

Model simulation for winter 2022/2023

Results（仿真结果-2022年12月）

- The **Combined scenario** experiences significant NSE, even during off-peak hours.



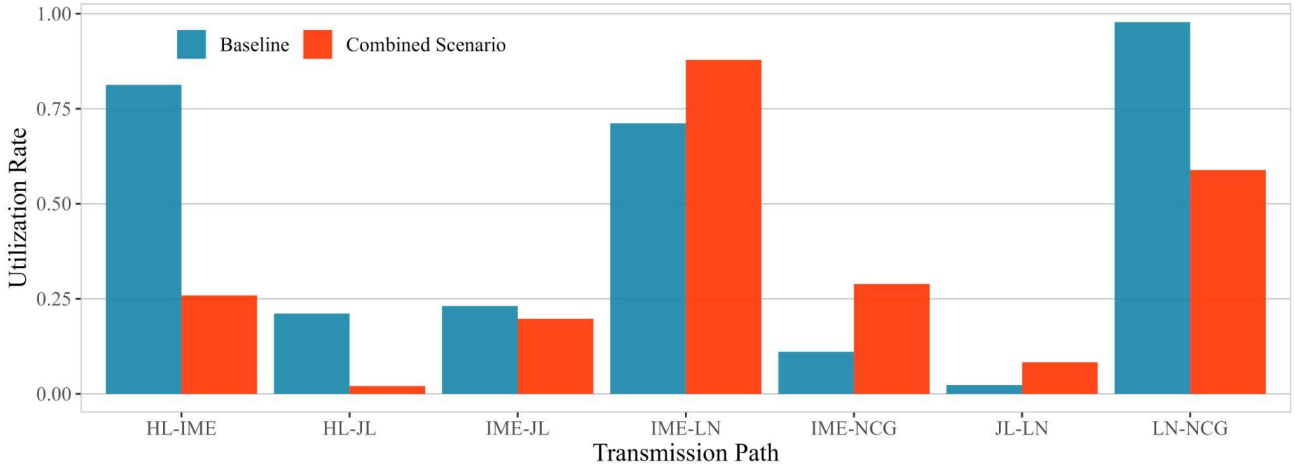
- Reducing CHP units' capacity dramatically compresses the NEG's capacity adequacy and peak-shaving flexibility during low wind and high load hours.

- 东北电网在**因素叠加情景**下经历严重的资源不充足，即使在非高峰时段也产生大量电力缺口。
- 热电联产机组最大出力的削减极大压缩了东北电网在风电供应不足以及高负荷时段的容量充足性以及削峰灵活性。

Model simulation for winter 2022/2023

Results（仿真结果-2022年12月）

- Average utilization rate of transmission paths in the **Baseline scenario** was 43%, while in the **Combined scenario** it was 33% .
- Decrease in utilization rate is due to energy availability constraints, leading to a trend of self-sufficiency in all provinces after fulfilling export obligations to NCG.
- Decreased support from Jilin to Heilongjiang in Combined Scenario results in decreased usage of the Jilin-Heilongjiang path and increased NSE in Heilongjiang.



- **基准情景**中输电线路的平均利用率为43%，而**因素叠加情景**的结果为33%。
- 利用率减少是由于能源供应的限制导致所有省份在履行对华北电网送电任务后都出现了自给自足的趋势。
- 在**因素叠加情景**下，吉林对黑龙江的支持不足导致吉林向黑龙江输送的电力减少，从而增加黑龙江的NSE。

Model simulation for winter 2022/2023

Results（结果对比-2022年12月&2023年2月）

- The results of the February 2023 simulation are consistent with the findings from December 2022. The main input difference is that February 2023 has lower average and maximum power demand.

Modeling Periods	NE Max. Demand (MW)	NE Avg. Demand (MW)	NCG Max. Demand (MW)	NCG Avg. Demand (MW)
Dec 2022	84,239	74,435	4,776	4,012
Feb 2023	78,383	70,146	4,124	3,658

- 东北电网资源充裕度在2023年2月的表现与其在2022年12月的表现相似，模型输入的主要差别在于2023年2月有着更低的最大和平均电力需求。

- Decline in power demand led to decreases in NSE and path utilization in extreme conditions, as well as more frequent storage charging and discharging.

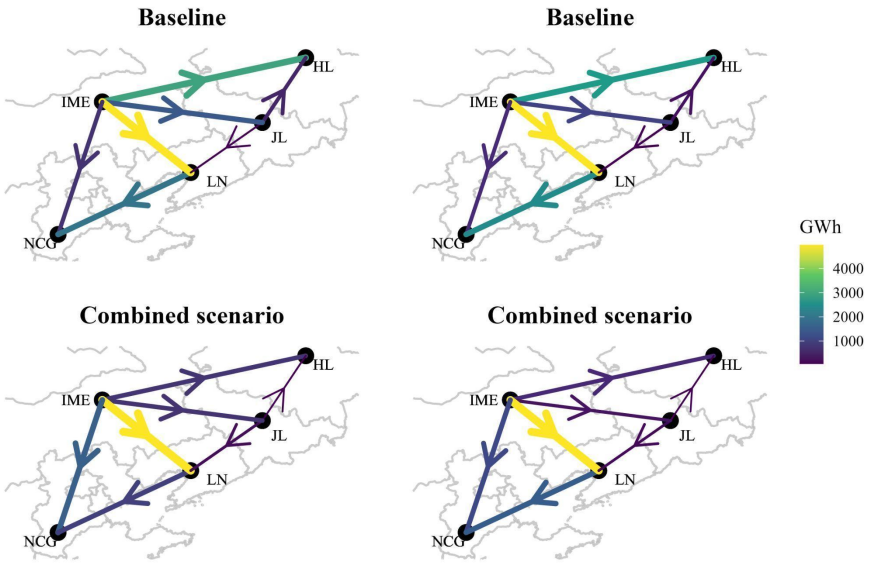
Modeling Periods	NE Max. NSE (% of Demand)	NE Avg. NSE (% of Demand)	Avg. Path Utilization (%)	Total Storage Charging (MWh)
Dec 2022	29.3	7.2	33.1	65,879
Feb 2023	26.0	6.1	31.0	151,133

- 电力需求的减少导致在因素叠加场景下更低的电力缺口以及输电线利用率。还使得储能设备更频繁的进行充放电以减缓系统压力。

Model simulation for winter 2022/2023

Results（结果对比-2022年12月&2023年2月）

- Liaoning usually imports power, but plays a significant role in exporting to NCG.
- In **Combined Scenario**, all provinces need more capacity locally, resulting in less inter-provincial support, while Liaoning still has significant export tasks due to export contracts, exacerbating the power crunch.
- Inner Mongolia East exports most power to other provinces, with Jilin assisting when wind power resources are sufficient, supporting Heilongjiang and Liaoning during normal conditions.



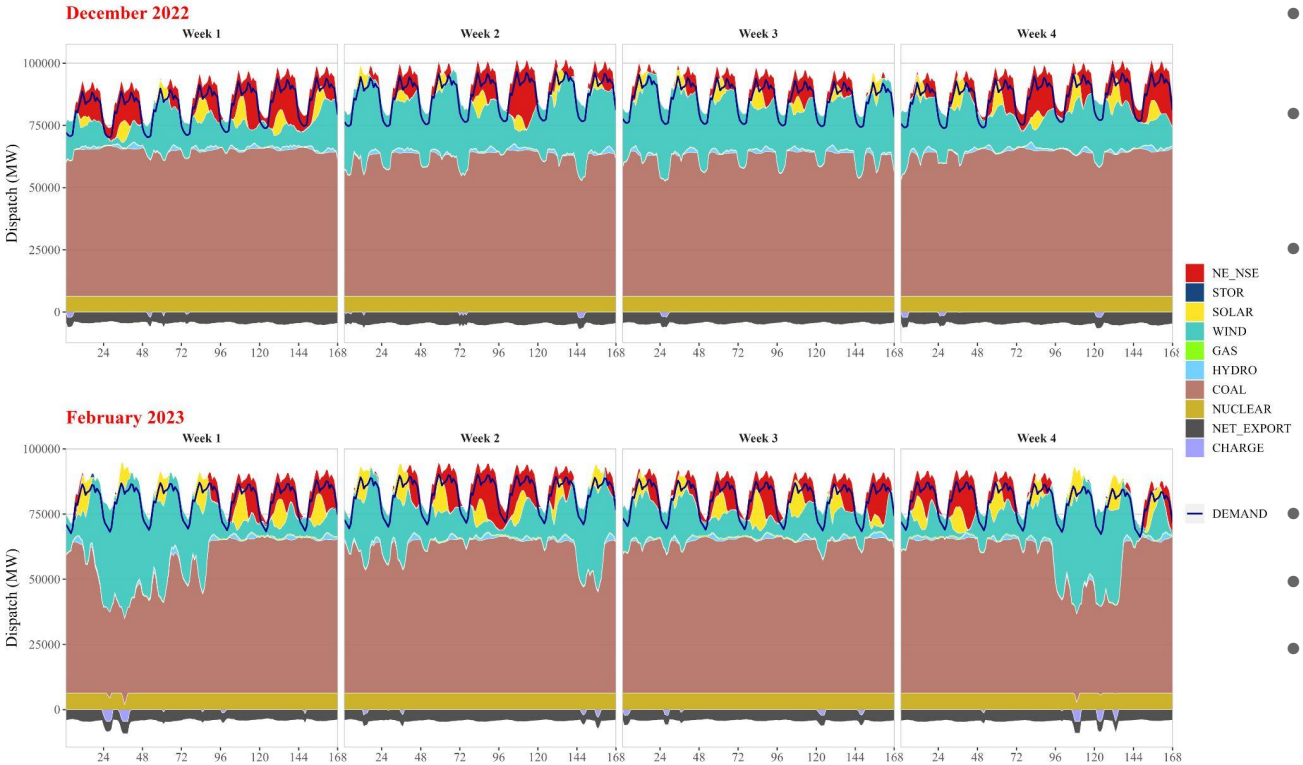
- 虽然辽宁总体上是一个净电力输入省，但它在向华北电网送电方面发挥着重要作用。
- 在**因素叠加情景**下，所有省份在本地都需要更多的 电力产能，导致省际支援减少，而辽宁因为送电合同义务仍需承担重要的送电任务，加剧了电力危机。
- 蒙东电网是向其他省份 输送最多电力的地区，而吉林在 风力资源充足时可以辅助支援外输，为黑龙江和辽宁提供支持。这在正常情况下是很有帮助的。

Figure: Transmission net flow (December 2022, left; February 2023, right)

左图：输电网络潮流 (左边为2022年12月情景，右边为2023年2月情景)

Model simulation for winter 2022/2023

Results（结果对比-2022年12月&2023年2月）



- The availability of wind power in NEG is correlated with the occurrence of NSE due to its high share in generation mix.
- AC lines within the region connecting Inner Mongolia East with Jilin and Heilongjiang improve resource adequacy and wind power balancing.
- Energy storage could help mitigate resource adequacy challenges, but NEG lacks sufficient storage capacity or have storage units that are not operated often in the system under pressure.

- 由于风电在发电组合中所占的比重较高，东北电网中风电的可用性与NSE的发生高度相关。
- 连接蒙东电网、吉林和黑龙江的交流线路可以有效提高资源充足性和风电平衡。
- 储能设备可以帮助缓解资源充足性挑战，但东北电网缺乏足够的储能容量或储能设备在系统压力下运行频率低。

International Experience

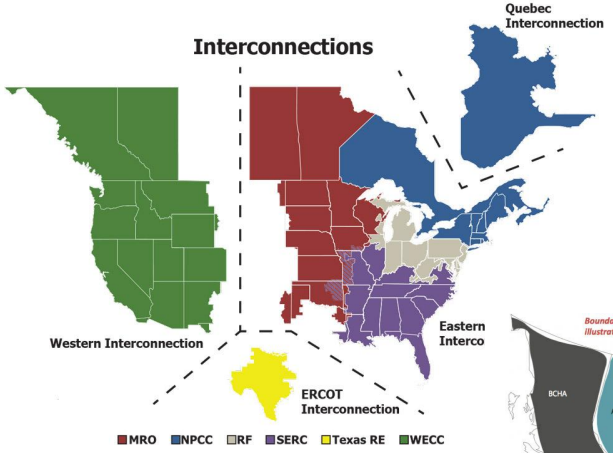
Resource Adequacy Challenges in California（加州的资源充足性挑战）

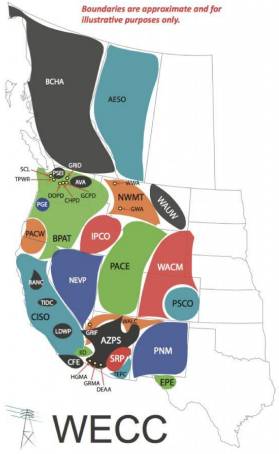
- Recent power shortage and tightness events:
  - In 2019, the power of over 3 million Californian customers was disrupted due to public safety power shut-offs (PSPS).
  - In 2020, rolling blackouts affected 0.5 million Californian customers.
  - In 2022, emergency alerts prompted unprecedented consumer reductions in demand, averting the need for more extreme measures.
- Contributing factors:
  - High demand (region-wide coherent extreme weather events).
  - Power supply availability challenges (renewable energy, generators offline).
  - Transmission network availability challenges (entire region facing similar challenges, decreasing intra-regional exports).
  - Wildfire risks lead to transmission line de-energization.

- 近期电力短缺和紧缩事件：
  - 2019年，由于公共安全停 电（PSPS），超过300万加利福尼亚客户的电力受到影响。
  - 2020年，滚动停电影响了50万加利福尼 亚客户。
  - 2022年，紧急警报促使消费者减少需求，避免了采取更极端的措施。
- 造成因素：
  - 高电力需求（区域范 围内连续极端天气事件）。
  - 电力供应可用性挑战（可再生能源，发电机停机）。
  - 输电网络可用性挑战（整个地区面 临类似的挑战，减少区域内的出口）。
  - 野火风险导致输电线路停电。

International Experience

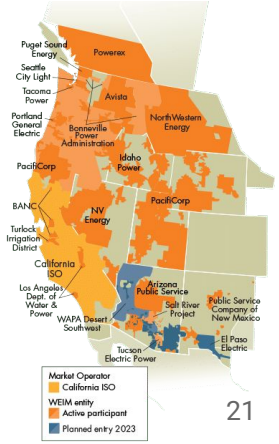
Western U.S. Energy Markets（美国西部的电力市场）





Source: Western Electricity Coordinating Council

- Regional transmission organizations (RTO) incorporate multiple functions, including:
  - Energy balancing and reserve markets
  - Resource adequacy planning
  - Transmission planning and coordination
- RTO-lite arrangements evolving in the absence of single market, including the Western Energy Imbalance Market:
  - Voluntary, utility-based opt-in
  - Real-time balancing
  - Moving to day-ahead option
- 区域输电组织（RTO）包括多个功能，包括：
  - 能源平衡和备用市场
  - 资源充足性规划
  - 输电规划和协调
- 西部能量不平衡市场，在缺乏单一市场的情况下逐步发展的RTO-lite模式：
  - 自愿、基于公用事 业的选择
  - 实时平衡
  - 逐步转向日前市场



International Experience

Resource Adequacy Challenges in California（加州的资源充足性挑战）

- Context:
  - Many IRP requirements on load-serving entities (LSEs)
  - Aggressive clean energy goals (100% by 2045)
  - Consumer adoption of distributed energy
  - Community choice aggregation
- Effects on markets
  - Wholesale prices go down
  - Out-of-market prices (e.g., reliability must-run, long-term contracts) go up
  - **Mismatch: short-run marginal ≠ long-run marginal costs**
- *Current (sub-optimal) direction:*
  - Governments enhance role in resource adequacy, including mandating flexibility resources
  - Spot market used for short-run efficiency

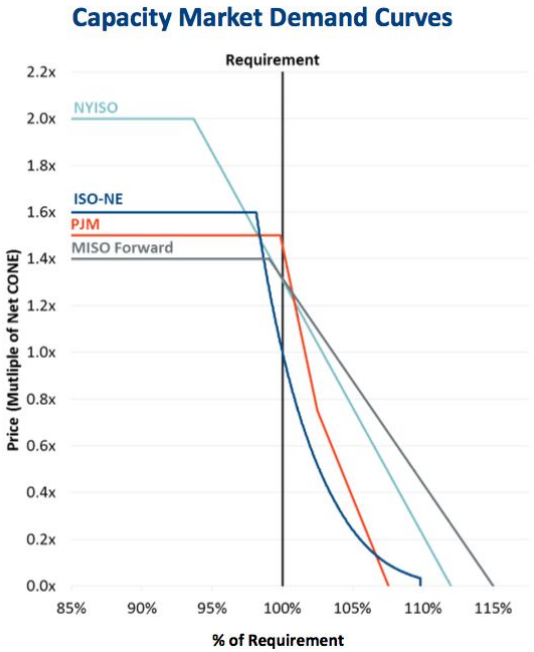
- 背景：
  - 对负载服务实体（LSEs）的许多IRP要求
  - 激进的清洁能源目标（到2045年实现100%）
  - 消费者采用分布式能源
  - 社区选择聚合
- 对市场的影响
  - 批发价格下降
  - 市场外价格（例如，可靠性必 须运行，长期合同）上升
  - **不匹配：短期 边际成本≠长期边际成本**
- *目前（次优）的方向：*
  - 政府加强资源充足性的角色，包括 强制要求灵活性 资源
  - 利用现货市场实现短期效率

(Tierney, 2018; Brattle, 2018)

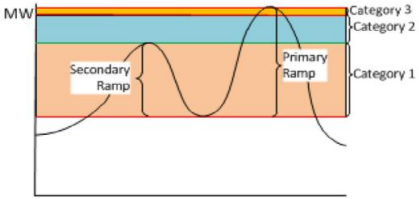
International Experience

Flexibility in the long-run（长远角度的系统灵活性）

Traditional Capacity Markets
传统容量市场



California: Flexible Resource Adequacy (Bilateral Contracts)
加州：灵活资源充足性（双边合同）



	Category 1	Category 2	Category 3
Must-offer obligation	17 Hours	5 Hours	5 Hours
	5 AM- 10 PM Daily For the whole year	3 PM to 8 PM for May – September	3 PM to 8 PM for May – September
	5 AM- 10 PM Daily For the whole year	2 PM- 7 PM for January- April and October-December	2 PM- 7 PM for January- April and October-December
	Daily	Daily	Non-holiday weekdays
Energy limitation	At least 6 Hours	At least 3 Hours	At least 3 Hours
Starts	The minimum of two starts per day or the number of starts feasible with minimum up and down time	At least one start per day	Minimum 5 starts a month
Percentage of LSE portfolio of flexible resources	At least 62 % for May – September	Up to 38% for categories 2 and 3 combined	Up to 5%
	At least 46 % for January- April and October-December	Up to 54% for categories 2 and 3 combined	Up to 5%

Policy Recommendations for China

Markets and prices（市场与价格视角）

- The 2021 Blackout in Northeast China was caused by rigidities in the administered coal on-grid electricity price mechanism.
- Coal prices increased significantly over the course of 2021, but the on-grid price was not flexible enough to reflect this surge, leading to withholding of coal generation capacity.
- We recommend expediting implementation of a unified regional Northeast spot market to rationalize compensation to resources and eliminate overly-restrictive price limits.
- The priority should be to create a time-varying wholesale spot price that reflects system conditions and offers reasonable compensation to resources.

- 2021年停电事故，主要是煤电上网电价机制僵化所致。
- 煤炭价格在 2021 年期间大幅上涨，但上网电价不够灵活，无法反映这一激增，导致煤电机组发电能力被抑制。
- 我们建议加快建立统一的东北区域现货市场，以合理补偿资源并取消过度限制的价格帽。
- 优先事项应是创建一个随时间变化的批发现货价格波动机制，以反映系统条件，为各种资源提供合理补偿。

Policy Recommendations for China

Planning processes（规划流程视角）

- Improved forward-looking analysis can help support the reliability and safety of electricity systems, particularly in marketized systems with growing shares of renewable energy.
- Planning processes need to be updated and improved to suit the characteristics of a competitive new energy system.
- We recommend the processes routinely publish forward-looking and backward-looking resource adequacy reports.
- This process should provide detailed scenarios, including release of data and methodological information, to complement the emerging electricity markets.

- 完善的前瞻性分析有助于支持电力系统的可靠性和安全性，尤其是在可再生能源份额不断增加的市场化体系中。
- 需要更新、改进和进一步规范规划流程，适应新型能源系统的特点。
- 项目流程应定期发布前瞻性和回顾性资源充足性报告。
- 该项目应提供详细的情景，包括发布数据和方法信息，以补充新兴的电力市场。

Policy Recommendations

Additional policy perspectives（其他政策视角）

- Resource use optimization is more important than thermal generation capacity
  - Dispatch practices, regulation, market rules, business models, and incentives, are crucial for ensuring power system reliability.
- Constructing additional coal-fired power plants is unlikely to be the least-cost answer to support a reliable, new energy power system.
  - A power system with high levels of flexibility can maximize the use of inexpensive variable renewable resources while optimizing the role of other resources in supporting roles.
  - In the long term, new "clean firm" power resources are needed for future electricity mixes with even higher penetrations.
- Continue developing policy and market rules to unlock the benefits of demand response and energy savings.
  - Incentivize holistic development of flexible resources, including demand side resources and energy saving.
  - Fast-responding resources like demand response, energy storage, and gas-fired capacity can provide valuable flexibility services and help integrate growing shares of renewable energy.
  - Resources should be fully and fairly rewarded and can provide alternatives to “orderly use” policies with large societal costs.

- 应强调优化资源利用，而不是仅仅专注于拥有足够的热发电能力。
  - 调度工作、政府规章制度、市场规则、商业模式和激励政策手段对于确保电力系统的可靠性也至关重要。
- 建设更多的燃煤电厂不太可能是支持可靠的新能源电力系统成本最低的解决方案。
  - 具有高度灵活性的电力系统可以最大限度地利用便宜的波动性可再生资源，同时优化其他资源的支撑作用。
  - 从长远来看，需要新的“清洁可调度能源”电力资源来支持未来可再生能源渗透率更高的电力结构。
- 必须继续出台政策，制定市场规则，以发挥需求响应和节能效益。
  - 鼓励全面开发更灵活的资源，包括需求侧资源和节能。
  - 需求响应、储能和燃气发电机组等快速响应资源可以提供有价值的灵活性，并有助于整合更多的可再生能源。
  - 资源应该得到全面和公平的回报，并且可以为社会成本昂贵的“有序用电”政策提供替代低成本方案。

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Thank you.

Appendix

附录

- The data utilized in this study were obtained solely from public sources, specifically the Global Energy Monitor (GEM), WIND Financial Terminal (WIND), Renewables Ninja (RN), PowerGenome (PG), as well as news articles and reports (NR). Further details regarding the sources of data can be found in Table A.1.

Table A.1 Data sources for generators information

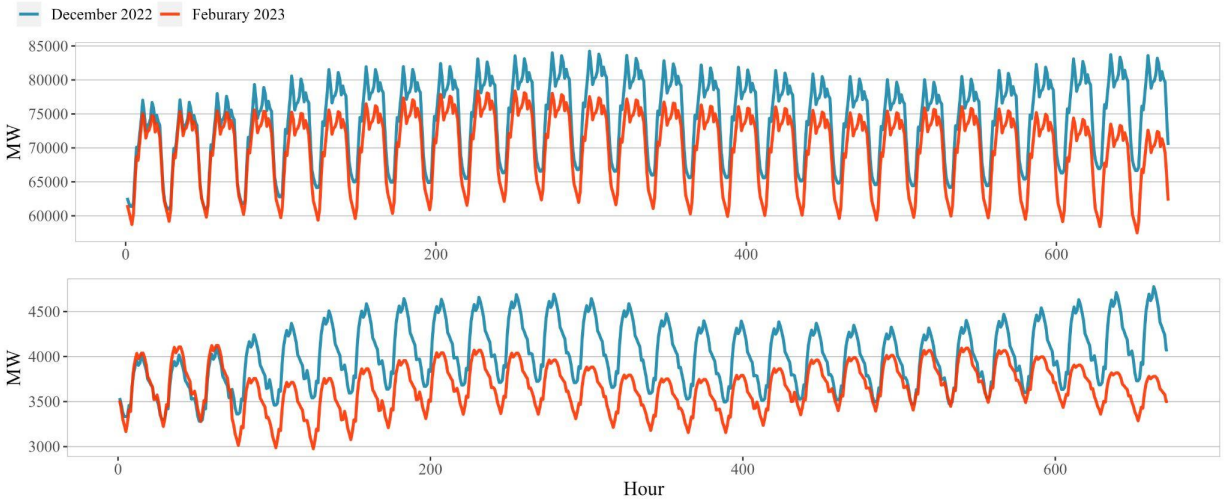
Generator Type	Capacity	Performance	Capacity factors	Heat Rate	Start cost	Fuel prices
Coal	GEM	NR	1	GEM	PG	WIND
Gas	GEM	NR	1	PG	PG	WIND
Wind	WIND	1	RN	0	0	0
Solar	WIND	1	RN	0	0	0
Hydro	WIND	PG	WIND	0	0	0
Nuclear	NR	NR	1	-	PG	0
Storage	NR	1	1	0	0	0

Appendix

附录

- Simulated power demand for NE and NE to NCG export obligations are shown in Figure A.1.

Figure A.1 Demand for Winter 2022 / 2023 (NE, top; NE to NCG export obligation, bottom)



Appendix

附录

- We show fuel prices in Table A.2. Coal prices in Baseline is calculated based on Qinhuangdao Q5500 annual long-term contract price + transportation cost to each zone (assume 0.33 RMB/(ton*km)). Coal prices in High Fuel Price/Derating Capacity refers to the monthly long-term coal prices + transportation cost calculated from:
  - Monthly long-term contract coal price (weekly index) = 1/3 * ((CCI index + CCTD index + CECI index) at last Tuesday)
- Gas prices in all scenarios refer to Natural Gas Benchmark Price published by the State Council, as shown in Table A.3.

Table A.2 Coal prices ($/MMBTu)

Provinces	Baseline Coal Price	High Coal Price
Heilongjiang	5.82	7.27
Inner Mongolia East	4.74	8.19
Jilin	5.38	8.84
Liaoning	4.52	7.98

Table A.3 Gas prices ($/MMBTu)

Provinces	Benchmark Gas Price
Heilongjiang	6.56
Inner Mongolia East	4.88
Jilin	6.56
Liaoning	7.36

Appendix

附录

- Transmission lines aggregated in paths and their voltages are illustrated in Figure A.2. Line losses incurred during transmission are calculated under parallel lines (R = 1 / sum_k(1/r_k) , r_k = resistance over line k, where r_k is indicated in terms of %) based on the specific losses in Table A.4.

Figure A.2 NE network

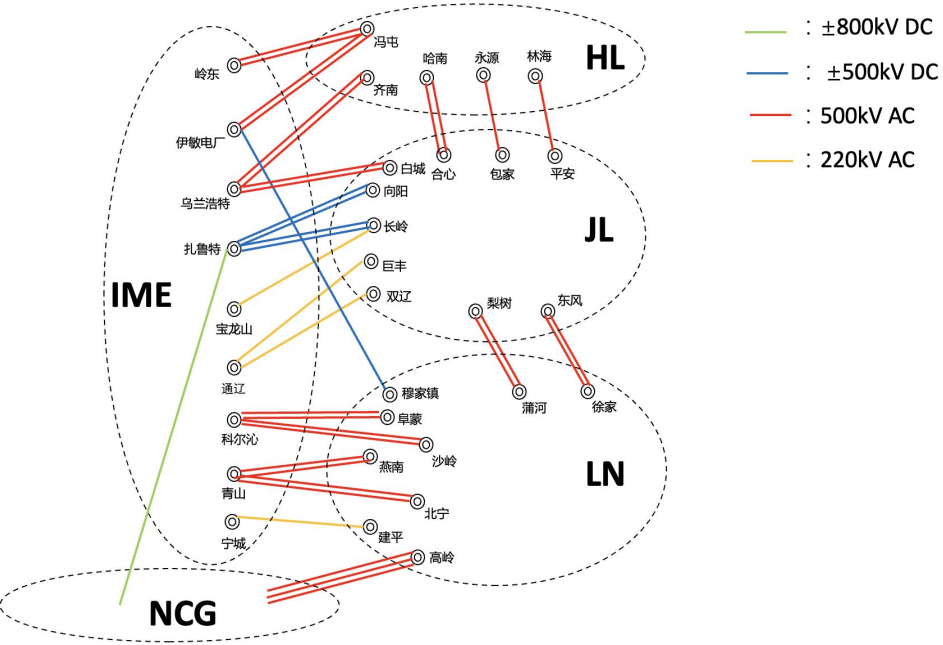


Table A.4 Transmission losses

Transmission Line	Losses (per 100 miles)	Transmission Line	Losses (per 100 miles)
±800 kV DC	0.31%	±500 kV DC	0.45%
500 kV AC	1.30%	200 kV AC	4.20%

Appendix

附录

- About two-thirds of all coal-fired installed capacities in NE are from CHP units, which provide 70% of NE's heat load in winter. When the units produce heat at its maximum, the corresponding maximum output for electricity will decrease as shown in Table A.5.

Table A.5 Maximum power output for electricity from CHP units

Rated Capacity (MW)	Reduced Capacity (MW)	Number of CHP units
0 - 150	No change	120
200	155	39
300	241	30
330	279	61
600	410	9
800	No change	2
1000	No change	2

Appendix

附录

- Other main assumptions:
  - Extreme Weather models a 15% increase in power demand across all zones, and uses the lowest average renewable energy profiles over the past 10 years for relevant months.
  - High Fuel Price and Derated Capacity simulates the conditions of the 2021 Northeast Blackout, with extreme fuel prices observed in October 2021 and a 20% reduction in total coal generator capacity from electricity-only coal generators. We derate coal generator capacities for each zone separately, starting from the least efficient units (the highest heat rate) until 20% of total coal capacity is reached.
  - Prioritized Export prioritizes power export to NCG by implementing a higher NSE penalty in NCG compared to NE, as stipulated by their long-term power transmission contract. Combined combines Extreme Weather, High Fuel Price and Derated Capacity, and Prioritized Export.
  - Unmet demand incurs $9,000/MWh for all zones and NCG in Baseline scenario. Prioritized Export reduces this number to $4,500/MWh for NE zones.
  - For each zone, the up reserve requirement consists of 5% of demand plus 10% of dispatch from renewable energy resources, and includes a contingency (the largest thermal unit or AC transmission line capacity, might be defined as N-1 contingency). We ignore operating reserves for NCG because NE only needs to meet NCG demand. Down reserve requirement is the same as up requirement, except that there is no contingency and non-dispatchable resources can contribute to down reserve.
  - Hydro inflow refers to historical hourly generation data divided by capacity (similar to capacity factor of non-dispatchable resources). Capacity and historical generation data are obtained from the WIND Financial Terminal, and generation is assumed to remain constant throughout the month.